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FAST TRACK PAPER

# Late Mesoproterozoic (*ca* 1.2 Ga) palaeomagnetism of the Albany–Fraser orogen: no pre-Rodinia Australia–Laurentia connection

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## SUMMARY

New palaeomagnetic data from metamorphic and mafic intrusive rocks of the Albany–Fraser orogen place Australia at high latitudes at *ca* 1.2 Ga, whereas existing data show that Laurentia occupied low latitudes at this time. No previous reconstructions linking Australia and Laurentia at *ca* 1.2 Ga are supported by our new data. The two continents moved independently during the early Grenville orogeny, although it is possible that late Mesoproterozoic convergence resulted in an AUSMEX configuration by 1070 Ma.

**Key words:** Albany–Fraser orogen, dolerite dykes, Mesoproterozoic, metamorphism, palaeomagnetism, Rodinia.

## 1 INTRODUCTION

Possible late Meso- to Neoproterozoic connections between Australia–Antarctica and Laurentia have featured prominently in recent debate regarding the configuration of the Rodinia supercontinent. Australia is placed adjacent to western Canada according to the SWEAT hypothesis (Moores 1991; Dalziel 1991; Hoffman 1991), whereas the AUSWUS model positions Australia against the southwest United States (Brookfield 1993; Karlstrom *et al.* 1999; Burrett & Berry 2000). Both reconstructions are based mainly on correlation of Archaean and Palaeoproterozoic blocks and orogenic belts across the inferred rifted margins of Laurentia and Australia–Antarctica, hence a fundamental tenet of these models is that a connection existed between these two continents since at least Palaeoproterozoic time.

Reliable 1070 Ma palaeomagnetic data place Australia at latitudes too low to allow SWEAT or AUSWUS, but permit an alternative fit, AUSMEX (Wingate *et al.* 2002a,b). AUSMEX suggests that the late Mesoproterozoic Cape River Province of NE Australia (Blewett *et al.* 1998) correlates with the Grenville Province of southernmost Laurentia (Mosher 1998), and therefore may also imply an older connection between the two continents. Wingate *et al.* (2002a), however, noted that a comparison of preliminary palaeomagnetic results for Australia at 1.2 Ga with Laurentian data argues strongly against any late Mesoproterozoic connection. In this paper, we present these new palaeomagnetic data for Australia, and compare them with Laurentian results to explore possible late Mesoproterozoic connections between the two continents.

## 2 GEOLOGY AND SAMPLING

The Albany–Fraser orogen (Fig. 1) records a two-stage collision of the Gawler–Mawson craton with the south and southeast Yilgarn craton (Harris 1995; Clark *et al.* 2000). Structures formed between 1345 and 1260 Ma by west-directed thrusting and dextral transpression along the Yilgarn margin were reactivated between 1215 and *ca* 1140 Ma during dextral transpression in the Albany mobile belt and northwest-directed thrusting and sinistral shearing along N–S shear zones in the Fraser mobile belt (Black *et al.* 1992; Harris 1995; Nelson *et al.* 1995; Clark *et al.* 2000). The Mount Barren Group, located at the junction of the Albany and Fraser mobile belts (Fig. 1), consists of Palaeoproterozoic metasedimentary rocks that rest nonconformably on, or are thrust over, Archaean basement (Harris 1995; Wetherley 1998), and were metamorphosed to lower greenschist to amphibolite facies at 1205 ± 10 Ma (Dawson *et al.* 2003). Orogen-parallel dolerite dyke swarms, dated at *ca* 1210 Ma, occur along the Yilgarn craton margins (Wingate *et al.* 2000).

Samples were obtained from three localities. Twenty-one samples of granulite facies rocks in the Bremer Bay and Whalebone Point areas of the Central Domain of the Albany mobile belt (Fig. 1b) were collected to augment previous work by Pisarevsky & Harris (2001), in which the geology of these sample sites is summarized. Fifty-one samples were collected from quartzites, quartz-mica schists, and shales of the Mount Barren Group (Fig. 1b). The same open pit mine exposure of the quartz dolerite Fraser dyke dated at 1212 ± 10 Ma by Wingate *et al.* (2000) was sampled in detail, together with its Archaean dolerite host rocks, for a baked contact test (Fig. 1c).



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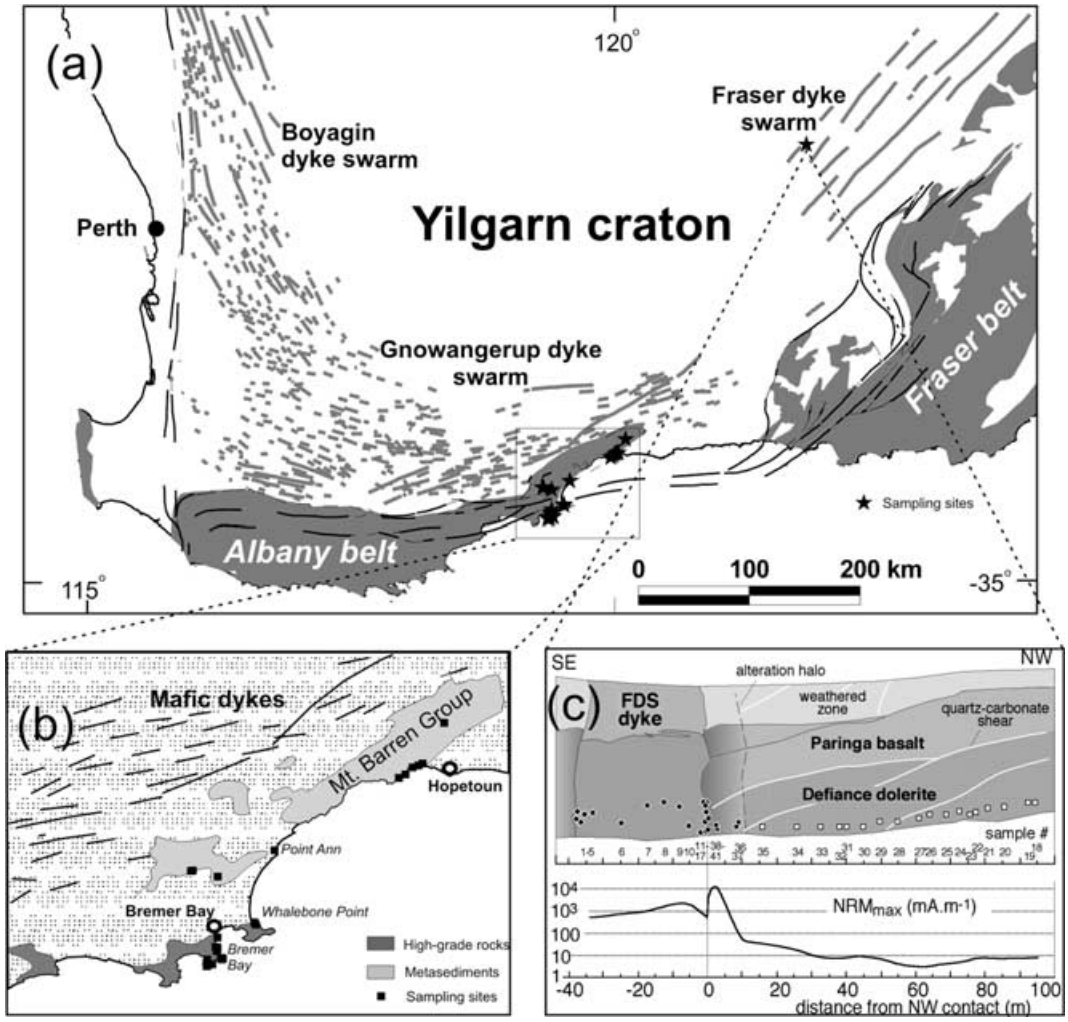



**Figure 1.** (a) Simplified geological map of the Albany–Fraser orogen, which consists of the Albany and Fraser mobile belts. (b) Bremer Bay and Mt Barren localities. (c) Forty-one samples were collected from a 1212 ± 10 Ma quartz dolerite dyke of the NE-trending Fraser swarm and its host Archaean dolerite exposed in the Defiance open pit at the Victory Gold Mine, Kambalda. The graph shows the variation in maximum magnetization intensity with distance from the dyke contact.

## 3 PALAEOMAGNETIC RESULTS

### Bremer Bay—Whalebone Point (BB1)

Demagnetization characteristics and rock magnetic experiments show that single domain (SD) and pseudo single domain (PSD) magnetite are the main carriers of stable remanence in the high-grade rocks (Fig. 2a), confirming previous results of Pisarevsky & Harris (2001). A steep bipolar component, BB1, isolated in 29 samples of tonalitic gneiss, mafic granulite, and metadolerite (Fig. 2a; Table 1) passes the reversal test of McFadden & McElhinny (1990) with classification $R_c$.

### Mount Barren Group (MBG)

Most quartzites yield a single stable steep remanence resembling BB1. Demagnetization curves (Fig. 2b) and rock magnetic experiments, including IRM acquisition and Lowrie's (1990) 3-axis method, show that the main magnetic carrier is haematite. This steep component, MBG, has been isolated in 30 samples (Fig. 2b; Table 1), and passes a reversal test with classification $R_c$. Lower metamorphic grade quartz-mica schists and shales, however, possess very weak ($<1$ mA m$^{-1}$) magnetizations and exhibit no stable remanence. Rapid decay in intensity of these samples during heating to 100–150 °C suggests the presence of goethite. Stable haematite in the metamorphosed rocks may have formed by dehydration of goethite during metamorphism at 1205 Ma.

### Fraser Dyke (FDS)

After removal of a low-stability, present Earth's field (PEF) overprint from most samples, all dyke samples exhibit a steep, downwards-directed component, FDS, carried by magnetite (Fig. 2c). Baked contact samples yield similar directions, very high intensities (Fig. 1c) and are single-component, with no trace of a PEF overprint, indicating that the contact rocks contain SD magnetite formed during baking by the adjacent dyke. Most country rock dolerite samples far from the dyke (Fig. 1c) possess a NW-upwards direction that is clearly different from the dyke and PEF directions (Fig. 2c). Preservation of both stable dyke and country rock directions indicates that the dyke carries a primary thermal remanent magnetization that dates from the time of intrusion at 1212 Ma.

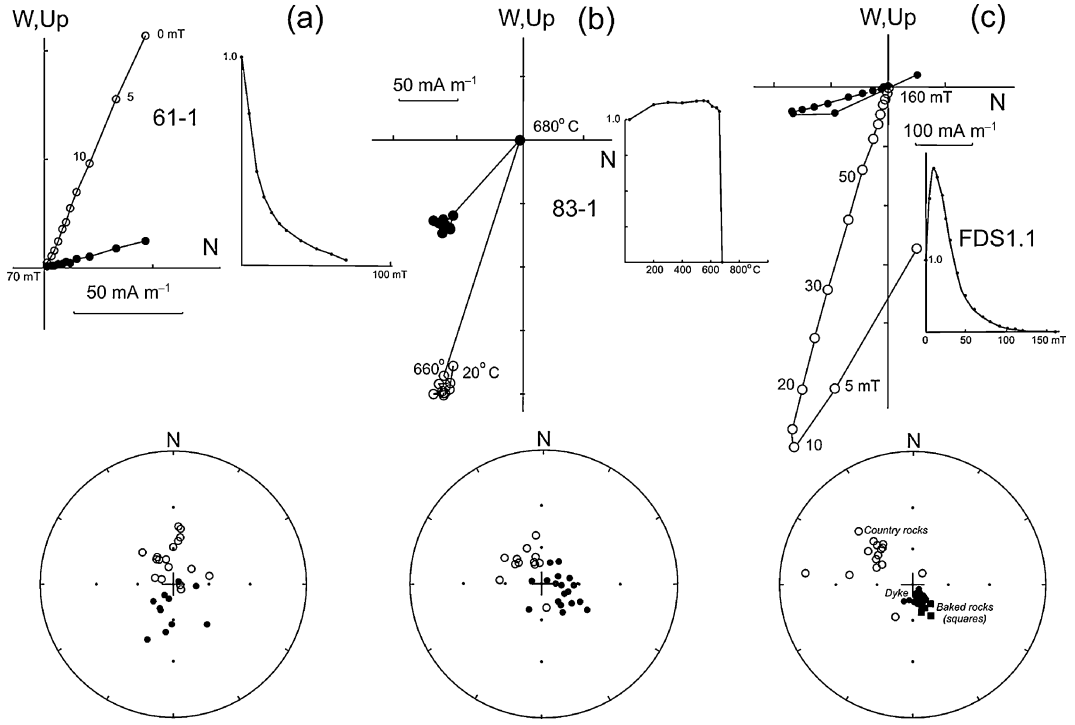



**Figure 2.** Stereoplots and examples of demagnetization behaviour for samples from (a) Bremer Bay—Whalebone Point metamorphic rocks, (b) Mount Barren metasediments, and (c) the Fraser dyke baked contact test. In orthogonal plots, open (closed) symbols show magnetization vector endpoints in the vertical (horizontal) plane; curves show changes in intensity during demagnetization. Stereoplots show upwards (downwards) pointing palaeomagnetic directions with open (closed) symbols.

**Table 1.** Mean directions and palaeopoles.

| # | Locality | Location | | | Mean direction | | | | Pole position | | | |
|---|---|---|---|---|---|---|---|---|---|---|---|---|
| | | Lat (°S) | Long (°E) | *B*/*N*/*n* | *D* (°) | *I* (°) | *k* | $\alpha_{95}$ (°) | Lat (°N) | Long (°E) | *dp* (°) | *dm* (°) |
| BB1 | Bremer Bay & Whalebone Pt. | 34.4 | 119.4 | 11*/29/58 | 178.2 | 67.0 | 27.4 | 8.9 | 74.4 | 303.8 | 12.2 | 14.7 |
| MBG | Mt. Barren Group | 34.0 | 119.8 | 12*/30/66 | 124.0 | 68.8 | 29.2 | 8.2 | 46.6 | 347.4 | 11.9 | 13.9 |
| FDS | Fraser dyke & baked contact | 31.3 | 121.8 | 1/23*/33 | 152.6 | 74.1 | 110 | 2.9 | 55.8 | 325.7 | 4.7 | 5.2 |

Notes: #, pole abbreviation; *B*/*N*/*n*, number of sites/samples/specimens; *D*, declination (E of N); *I*, inclination (positive downwards); *k*, Fisher's precision parameter; $\alpha_{95}$, circle of 95 per cent confidence about mean direction; *dp*, *dm*, semi-axes of cone of 95 per cent confidence about palaeopole.
*The level of statistics.

## 4 DISCUSSION

The Mount Barren Group and Bremer Bay palaeopoles (MBG and BB1) represent magnetizations acquired during cooling following the second stage of Albany–Fraser tectonism at *ca* 1.2 Ga. This is the main tectonothermal event recorded in the Albany Mobile Belt (Black *et al.* 1992) and is coeval with the second Mesoproterozoic event in the Fraser Mobile Belt (Harris 1995) dated at 1205 ± 10 Ma by Dawson *et al.* (2003). The consistency in palaeomagnetic directions determined in samples from different sites in the Mount Barren Group indicates that their magnetization postdates regional NW-verging thrusting and folding (Wetherley 1998). It is likely that the MBG pole is slightly older than the BB1 pole, because the magnetic blocking temperature of hematite is higher than that of magnetite. Proximity of the BB1 and MBG poles to the FDS pole for the 1212 ± 10 Ma Fraser dyke, and their distance from the 1140 Ma IAR pole (Fig. 3), indicates that the MBG and BB1 poles are similar in age to the dyke. Although well-dated, the FDS result represents a single spot reading of the Earth's magnetic field, and is not likely to have averaged palaeosecular variation (PSV). The BB1 and MBG poles, however, are based on antipodal directions and are likely to have adequately averaged PSV. Collectively, the three poles (MBG, BB1, and FDS) reliably indicate the position of Australia at about 1200 Ma (Fig. 3).

Laurentian palaeopoles between 1267 and ~1000 Ma form the so-called Logan loop (Fig. 3). The apparent polar wander (APW) path for Australia (excluding unreliable poles from central Australia—see discussion by Wingate *et al.* 2002b), has a very different shape and length (Fig. 3). This difference argues strongly against the existence of a combined Laurentia—Australia–Antarctica block since the Palaeoproterozoic, as implied by previous reconstructions.

In addition, in each of the three previous hypotheses (SWEAT, AUSWUS, and AUSMEX), the *ca* 1200 Ma palaeopoles for Australia are located 60° to 90° from those of Laurentia (Figs 4a–c), indicating that none of these scenarios existed at this time.

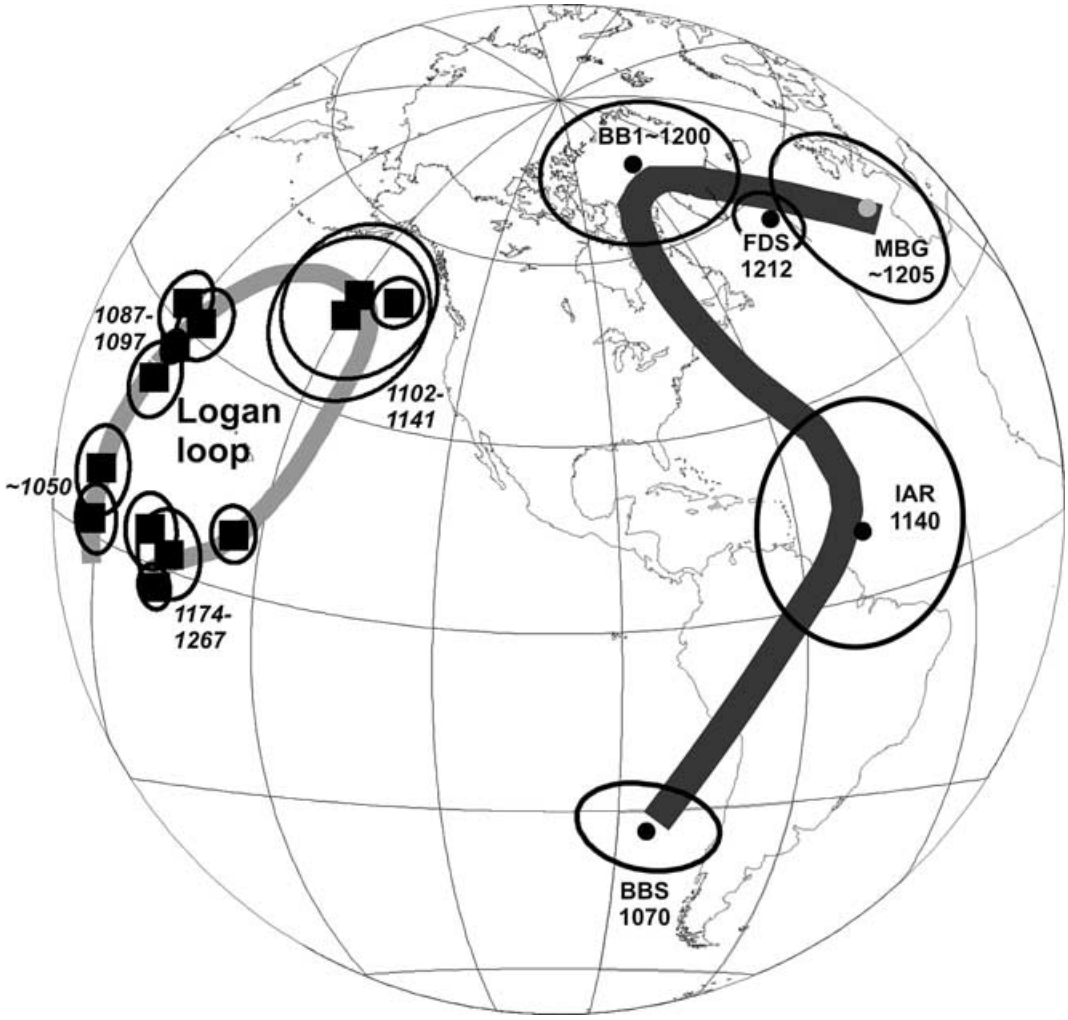



**Figure 3.** Late Mesoproterozoic apparent polar wander (APW) paths for Australia (black) and Laurentia (grey), in present coordinates. Laurentian data are summarized by Powell *et al.* (2001) and Tohver *et al.* (2002). Australian data include poles from this study (MGB, FDS and BB1), BBS (Wingate *et al.* 2002a), and IAR (Tanaka & Idnurm 1994). Ages are in Ma.

Imprecise poles for the Abitibi dykes (AB, Ernst & Buchan 1993) and Lakeview Dolerite (IAR, Tanaka & Idnurm 1994) overlap in all three reconstructions and indicate broadly that a connection between Australia and Laurentia is permissible by 1140 Ma.

Superimposing groups of coeval palaeopoles for Australia and Laurentia places the continents at approximately their correct orientations and palaeolatitudes and allows their possible relative positions to be explored (Figs 4d–f ). Matching poles at both 1200 Ma and 1140 Ma yields the closest arrangement of the continents, with northern Australia facing the western Laurentian margin (Fig. 4d). This scenario requires more than 90° of CCW rotation to bring the continents into any previously proposed Rodinia configuration. Another possibility is that Australia was separated from Laurentia by a wide polar ocean at *ca* 1200 Ma (Fig. 4e). Australia could reach a Rodinia fit with Laurentia, in this case, by closure of the intervening ocean and with minimal rotation. Note that longitudinal uncertainties allow a spectrum of possibilities. The alternate polarity option (Fig. 4f) is less likely because Australia would traverse the equator, rather than occupy high latitudes at 1140 Ma, as required by the IAR pole. In addition, the shortest convergence path shown in Fig. 4(f) does not leave sufficient space for proposed collisions of eastern Laurentia with Baltica and Amazonia at about the same time (e.g. Park 1992; Starmer 1996; Pisarevsky *et al.* 2003).

These scenarios assume that Australia was joined to Laurentia at 1070 Ma, although current palaeomagnetic data permit the absence of any late Mesoproterozoic connection between the two continents. Although evidence exists for a possible late Mesoproterozoic collision in the Cape River Province of northeast Australia (Blewett *et al.* 1998), a connection between the Cape River—Musgrave—Albany–Fraser and Grenville provinces (e.g. Karlstrom *et al.* 2001; Wingate *et al.* 2002b) is unlikely, because pre-1140 Ma palaeomagnetic data indicate that Australia and Laurentia were in relative motion, and probably far apart, during the early Grenville orogeny.

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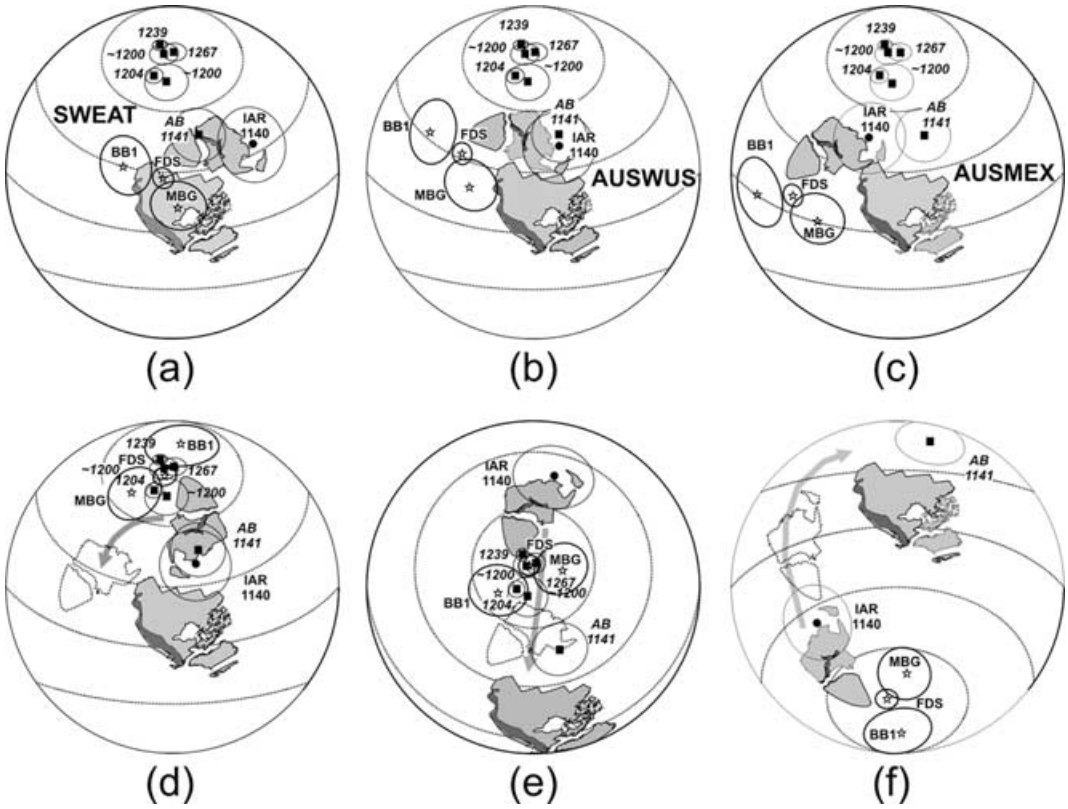



**Figure 4.** Australia-Laurentia reconstructions and their palaeomagnetic constraints: (a) SWEAT (Moores 1991; Dalziel 1991; Hoffman 1991); (b) AUSWUS (Brookfield 1993; Karlstrom *et al.* 1999; Burrett & Berry 2000); (c) AUSMEX (Wingate *et al.* 2002a); (d)–(f) three possible scenarios for Australia-Laurentia convergence between ~1260–1200 Ma and 1070 Ma, as discussed in the text. Australian poles: stars, new poles from this study; circle, IAR pole. Laurentian poles (squares): AB, Abitibi dykes (Ernst & Buchan 1993); see also caption to Fig. 3. Rotation parameters (Euler pole latitude, longitude, rotation angle): Laurentia to absolute framework (a)–(f): 38.2, −124.7, −119.0; Australia to Laurentia: (a) 32.5, 122.4, 130.7; (b) 49.9, 115.7, 114.4; (c) 54.5, 149.9, 120.2; (d) 40.0, 47.7, 96.5; (e) 8.8, 327.6, −104.2; (f) 52.3, 221.5, 177.4.

comments. This is Tectonics Special Research Centre publication #234, and a contribution to IGCP 440.

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